Object-Oriented Key Point Vector Distance for Binary Land Cover Change Detection Using VHR Remote Sensing Images

Zhiyong Lv, Tongfei Liu, and Jón Atli Benediktsson, *Fellow, IEEE*

***Abstract*— Very high-resolution (VHR) remote sensing images can geometrically depict ground targets in detail but are usually insufficient in the spectral domain. This characteristic leads to a considerable amount of noise and pseudo change in the produced binary change detection maps (BCDMs) when VHR remote sensing images are used for change detection. Here, to solve the aforementioned problem, an object-oriented key point vector distance (KPVD) is proposed to measure the change magnitude between bitemporal VHR images when land cover changes are detected. The proposed KPVD-based change detection approach comprises the following major steps. First, multiscale objects based on a postevent image are extracted by the fractional net evaluation segmentation approach, and then, the segments are taken as the unit for measuring the change magnitude between bitemporal images. Second, key points and the corresponding vector are defined to describe the object feature instead of using the total pixels within the object. Finally, KPVD is proposed to measure the change magnitude between the local areas referenced to the object in the bitemporal images. The change magnitude image (CMI) between the bitemporal images is generated while the entire images are scanned and processed object by object. A well-known automatic binary method, the Otsu approach, is employed in this article to divide CMI into a BCDM. Experimental results conducted on four real data sets demonstrate the feasibility and outperformance of the proposed KPVD-based change detection approach compared with five state-of-the-art methods in terms of visual performance and quantitative measurements.**

***Index Terms*— Key point vector distance (KPVD), land cover change detection (LCCD), multiscale segmentation, very high-resolution (VHR) remote sensing image.**

I. Introduction

LAND cover change detection (LCCD) with remote sensing images is defined as the process of finding the land cover change by comparing the bitemporal images that cover the same geographical area at different dates [1]–[3]. Due to the constantly changing surface of the Earth, LCCD plays an important role in many applications, such as landslide inventory mapping [4], [5], building change detection [6], [7], and even environmental protection [8]. Recently, with the increasing acquisition of remote sensing images with very high resolution (VHR), LCCD with VHR remote sensing images has been widely extended to urban development and planning [9], [10], geological disaster monitoring [11]–[13], and Earth resource management [14], [15].

However, due to the obtained bitemporal VHR images being acquired at different dates, the varying atmospheric conditions of the acquisition, such as sun height, soil moisture, and seasonal phenology, pseudo changes can be observed in the results of LCCD [16]–[18]. Furthermore, although VHR remote sensing images can geometrically capture the details of ground targets, they are insufficient in the spectral domain. This insufficiency leads to the "mixed pixel" phenomenon between different targets and reduces their separability [19], [20]. Therefore, reducing pseudo changes, maintaining the real change, and preserving the boundary among targets when using VHR remote sensing images remain a challenge in LCCD [21].

Many approaches have been developed for solving the aforementioned problems [7], [22]. One of the most popular methods is the spatial context-based approach. For example, Celik [23] exploited the contextual information through "$h \times h$" image block and then used principal component analysis and k-means clustering to further reduce the speckle noise of the LCCD map. In addition, sliding a regular window on the entire image for considering the contextual information around each pixel is an effective way for smoothing noise of the detection map [24]–[26]. However, a regular window usually blurred the boundary among different targets because the shape, size, and location of a target remain unpredicted in an image scene [27]. Thus, adaptive region extension is another way for reducing the noise and improving the accuracies of LCCD with VHR images to preserve boundaries while smoothing the noise in the detection map [16], [28]. In addition, fuzzy set theory and mathematical models are also often preferred for exploiting spatial-contextual information for LCCD. Liu *et al.* [29] proposed a change detection approach using change vector analysis and fuzzy c-means (CVA_FCM). Bazi *et al.* [30] introduced a multiresolution level set (MLS) method to find the global optimal contour for multispectral remote sensing

Manuscript received September 14, 2019; revised December 5, 2019 and January 21, 2020; accepted February 25, 2020. Date of publication March 13, 2020; date of current version August 28, 2020. This work was supported in part by the National Science Foundation China under Grant 61701396 and in part by the Natural Science Foundation of Shaanxi Province under Grant 2017JQ4006. *(Corresponding author: Zhiyong Lv.)*

Zhiyong Lv and Tongfei Liu are with the School of Computer Science and Engineering, Xi'an University of Technology, Xi'an 710048, China (e-mail: lvzhiyong_fly@hotmail.com; liutongfei_home@hotmail.com).

Jón Atli Benediktsson is with the Faculty of Electrical and Computer Engineering, University of Iceland, IS 107 Reykjavik, Iceland (e-mail: benedikt@hi.is).

Color versions of one or more of the figures in this article are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TGRS.2020.2977248

0196-2892 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:21:44 UTC from IEEE Xplore. Restrictions apply.

image change detection. Zhang *et al.* [31] proposed an unsupervised LCCD method based on level set evolution with local uncertainty constraints (LSELUCs). Despite numerous spatial context-based LCCD approaches have been developed, several studies have demonstrated that none of these methods is sufficiently universal and most of them are impractical [17], [18]. Therefore, the LCCD method can be improved in terms of accuracy and usability.

Apart from the spatial context-based approaches, due to the advantages of the object-based techniques for processing the VHR remote sensing images [32], [33], object-oriented change detection (OOCD) approaches have been applied successfully in practical applications [22], [34], [35]. Before applying the OOCD methods, multiscale segmentation is required to obtain "objects," which comprise a group of pixels with similar spectra and spatial continuity. The change magnitude between the pairwise objects is measured, and land cover change is achieved object by object [36], [37]. For example, Zhang *et al.* [38] proposed an OOCD based on the points of a box-whisker plot for LCCD with VHR remote sensing images. Cai *et al.* [4] promoted an OOCD approach for LCCD by adaptively fusing the advantages of multimethods (MAC_OOCD) to further improve the performance of OOCD methods. Cao *et al.* [39] promoted an approach by means of level set evolution and support vector machine (MLS_SVM), which combined the pixel-level and object-level detection results to output the final detection map. Although OOCD methods extend the feature dimensions for availability and play an important role in LCCD with VHR remote sensing images, few approaches focus on the definition of a measurement metric for the change magnitude between the pairwise objects. Therefore, developing an object-based distance for LCCD with VHR remote sensing images is important.

In this article, an object-oriented key point vector distance (KPVD) is proposed to measure the change magnitude between the pairwise local areas of an object for generating CMI to improve the LCCD performance and accuracies while using VHR remote sensing images. The proposed KPVD-based change detection approach has two objectives: 1) to develop a novel distance that is effective in resisting the bitemporal image variation in phenological season and imaging condition and 2) to construct a simple yet effective LCCD approach for detection land cover change with VHR remote sensing images.

The following steps are conducted to achieve the objectives. First, multiscale objects based on the postevent image are obtained by the fractional net evaluation approach (FNEA) segmentation algorithm [40], [41]. Second, with the benefit of box-whisper plot, key points of each object are extracted from the bitemporal image object by object. Finally, KPVD is defined based on the pairwise key point vectors (KPVs). While the entire object set is scanned and calculated in this manner, the CMI between the bitemporal images can be acquired. A widely used binary threshold, Otsu, is utilized in this article to divide the CMI into binary change detection map (BCDM). The proposed approach is applied to detect land cover change with four pairs of VHR remote sensing images and compared with the state-of-the-art and widely used methods, namely, CVA_FCM [29], MLS [30], LSELUC [31], MLS_SVM [39], and MAC_OOCD [4], to assess its feasibility and superiority.

The rest of this article is organized as follows. In Section II, the proposed KPVD-based change detection approach is comprehensively described. The experimental results are given and analyzed in Section III. Conclusions are given in Section IV.

II. Methodology

The proposed object-oriented LCCD approach measures the change magnitude between pairwise objects through a novel KPVD instead of directly using the traditional distance between the objects. Fig. 1 shows that the proposed KPVD-based approach is comprised of three steps: 1) a multiscale segmentation algorithm, FNEA, is applied to the postevent image for obtaining the multiscale object sets; 2) "key points" of an object of a specific temporal image are defined and extracted for the bitemporal images; and 3) the band-to-band distance between KPVs is proposed to measure the change magnitude between the pairwise areas of the object. Finally, the Otsu binary threshold method is adopted to separate the CMI into BCDM. The details of the proposed KPVD-based approach are provided in Sections II-A–II-D.

A. Brief Review of FNEA

Processing VHR images based on the multiscale objects can preserve the boundary of different targets and smooth the salt-and-pepper noise of results [32], [42]. Although the multiscale segments of the bitemporal images for the same target are usually different in terms of shape, size, and quantity, different segment sets from the bitemporal images can notably depict and preserve the boundary of the target. Therefore, the multiscale segments based on the postevent image for a target are assumed to depict the shape and size of the target in the pre-event image when the target is unchanged in the bitemporal images. Under this assumption, in this article, the multiscale segments based on the postevent image are taken as the "processing unit" to avoid the inconsistency between the segmentation of the bitemporal images.

In addition, the FNEA algorithm has the following advantages in the segmentation: 1) FNEA can produce boundary-constrained multiscale segmentation although the rule is not publicly known [43] and 2) it can be conveniently utilized because it has been embedded in the business image processing software eCognition as a tool for multiscale segmentation [44]. The FNEA [40], [41] multiscale segmentation approach is employed in this article to acquire multiscale objects. The FNEA algorithm generates the multiscale objects starting from pixels to objects by applying region merging and then fusing small objects into a large object. The FENA algorithm must be triggered by three parameters, namely, *scale*, *shape*, and *compactness*, during this region merging, the *scale* parameter decides the maximum allowed heterogeneity for the image objects, the *shape* determines the aim of the entire homogeneity criterion, and *compactness* is used to optimize image objects. Additional details regarding FNEA can be tracked in [45] and [46].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:21:44 UTC from IEEE Xplore. Restrictions apply.

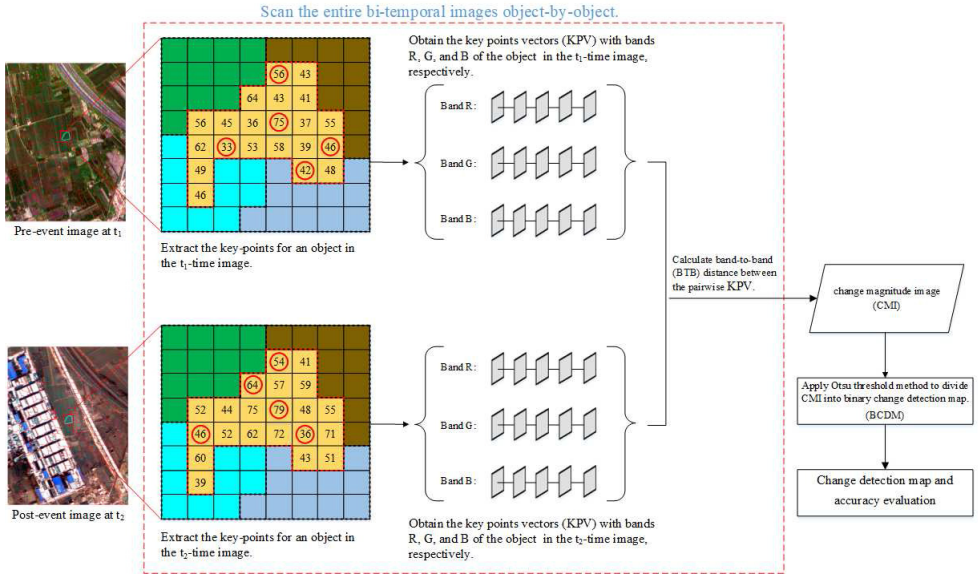


Fig. 1. Flowchart of the proposed KPVD-based approach for binary change detection.

B. Definition of Key Points for an Object

"Object" as referred to here comprises a ground of pixels that are spectrally homogeneous and spatially continuous. Thus, from an intuitive perspective, the first proposition in this article is to sparsely represent and describe the feature of an object. Only, a set of interest pixels is employed to characterize the feature of an object and not consider all its pixels. These pixels of interest within an object are called "key points" in this article.

Inspired by the pointwise graph-based local texture feature [47], describing the local feature of VHR image sparsely is helpful for reducing the negative affect from the outliers pixels [47]. Therefore, sparse description of the object feature using the key points of an object is proposed for LCCD in this article. The key points in this article are defined as the five points: k_0, k_1, k_2, k_3, and k_4. These points refer to different level spectral values of the pixels within an object. k_0 denotes the minimum spectral value, k_1 is the lower quartile spectral values, k_2 is the median spectral value, k_3 is equal to the upper quartile value, and k_4 is the maximum spectral value. A schematic example is displayed in Fig. 2 to illustrate the definition of key points for an object.

From the definition of key points of an object, key points can instead use the total pixels of an object due to the following issues: 1) the key points contain the "minimum" and the "maximum" spectral value, which depict the range of the pixels within the object and 2) according to the theory of descriptive statistics, quartiles based on the box-whisker plot well indicate the degree of dispersion and skewness in the data set. Therefore, key points can effectively express the feature of an object sparsely.

C. Object-Oriented KPVD

Based on the definition of key points, the feature of an object can be considered the KPV of the object. The KPVs of the

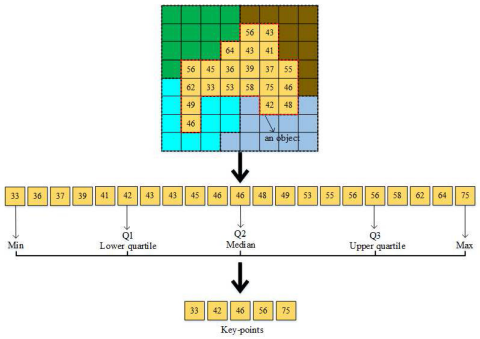


Fig. 2. Schematic example of extraction key points for an object.

object $o_i^{t_1}$ are assigned as $\mathrm{KPV}_i^{t_1}(r)$, $\mathrm{KPV}_i^{t_1}(g)$, and $\mathrm{KPV}_i^{t_1}(b)$ for the red, green, and blue bands, respectively, to measure the change magnitude between the two local areas ($o_i^{t_1}$ and $o_i^{t_2}$) of an object in the bitemporal images, while $KPV_i^{t_2}(r)$, $\mathrm{KPV}_i^{t_2}(g)$, and $KPV_i^{t_2}(b)$ represent the same information for $o_i^{t_2}$. Based on the inspiration of histogram-based bin-2-bin distance [48], [49], an object-oriented KPVD is proposed for measuring the change magnitude between the object areas in the bitemporal images. KPVD is defined as (1). In (1), if the distance between $\mathrm{KPV}_i^{t_1}(b)$ and $\mathrm{KPV}_i^{t_2}(b)$ is measured element by element, then (2) gives the same value as (1)

$$\Delta d_i = \frac{1}{2\times 3}\sum_{l=1}^{l=3}\left\|\mathrm{KPV}_i^{t_1}(l) - \mathrm{KPV}_i^{t_2}(l)\right\| \tag{1}$$

$$\Delta d_i = \frac{1}{2\times 3\times 5}\sum_{l=1}^{l=3}\sum_{j=0}^{j=4}\left\|k_j^{t_1}(l) - k_j^{t_2}(l)\right\| \tag{2}$$

where Δd_i is the change magnitude between the pairwise local area ($o_i^{t_1}$ and $o_i^{t_2}$) of the object o_i. In addition, l denotes a specific band of VHR remote sensing image with false color, and j represents an element within a KPV. According

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:21:44 UTC from IEEE Xplore. Restrictions apply.

to (1) and (2), the change magnitude Δd_i between the pairwise areas $o_i^{t_1}$ and $o_i^{t_2}$ can be obtained.

Based on the above definition, the proposed KPVD measures the change magnitude between the object area of the bitemporal images in two aspects.

1) The change magnitude is the average change magnitude between each band of the bitemporal images for a specific object area. This magnitude is more helpful for preserving information compared with transforming false-color images into gray images in traditional LCCD methods.
2) The proposed KPVD based on the sparse key points is more effective than directly using the total pixels in the band-to-band or the pixel-to-pixel distance for inhibiting the pseudo change for LCCD.

Due to the presence of outlier pixels in the total pixels, the negative influence of these outliers on the change magnitude measurement will be weakened by the proposed KPVD. Intuitively, key points are collected by the proposed BP, which can exclude the outliers from a data set. Moreover, one advantage of using KPVD is that the collected key points are nonparametric. Hence, these key points can display variation in samples of an object without assuming any underlying statistical distribution. KPVD is inspired by the histogram bin-2-bin distance [48], [49], but it measures the distance among pairwise objects individually using the key points of each object. Although basic spectral bands are adopted as the inputting feature in (1) and (2), KPVD is not limited in using this basic spectral feature.

Based on the above analysis, a large Δd_i value indicates a far distance between the bitemporal image in the object area. In other words, a large Δd_i denotes a large change magnitude between the object of the bitemporal images. By contrast, a small value of Δd_i implies a low possibility of change. When the entire bitemporal image is scanned object by object and computed by (2), then, CMI $= \{\Delta d_0, \Delta d_1, \Delta d_2, \ldots, \Delta d_m\}$, where m is the total number of the multiscale objects.

D. Changed/Unchanged Area Detection With Binary Threshold

Similar to many LCCD methods, a binary threshold is required to divide the CMI into BCDM. A widely used automatic binary threshold method, Otsu, is applied to divide the CMI into BCDM in many existing approaches [30], [48], [50]; Otsu can automatically obtain a global optimal threshold by maximizing the interclass variance between the two classes while minimizing the intraclass variance. Therefore, Otsu is used to separate the CMI into BCDM in the proposed KPVD-based change detection approach.

III. EXPERIMENTS

Before experiments, the preprocessing steps, which include co-registration and radiometric normalization, are required to minimize the difference among bitemporal images for LCCD. In our experiment, the bitemporal images are registered mutually using ground control points with the Georeferencing tool of ArcMap software. Radiometric normalization is achieved by the Radiometric Calibration function of ENVI 5.1 software. More details about co-registration and radiometric normalization can be tracked in [51] and [52].

TABLE I
DESCRIPTION OF THE DATA SETS IN THE EXPERIMENTS

Dataset	Size (pixels)	Ground Reference Map (changed/unchanged pixels)
A	923×593	$265, 93/520, 746$
B	750×950	$350, 66/677, 434$
C	768×1024	$505, 39/735, 893$
D	950×1250	$200, 483/987, 017$

The proposed KPVD was applied for LCCD with VHR remote sensing images to verify its effect and performance. Four pairs of VHR images referring to two kinds of real land-cover change events are employed for the evaluation. In addition, five state-of-the-art LCCD approaches, namely, CVA_FCM [29], MLS [30], LSELUC [31], MLS_SVM [39], and MAC_OOCD [4], were compared with the proposed KPVD-based change detection approach. Details will be provided in Sections III-A–III-D.

A. Data Description

Four pairs of image data sets that refer to the real landslide and land-use change events were used to test the feasibility and the performance of the proposed KPVD-based change detection approach, as shown in Fig. 3. In addition, the ground reference map of each data set was manually interpreted with the assistance of the Editing toolbar in the ArcMap 10.2 software. The details of the ground reference map are provided in Table I.

Data sets A, B, and C are aerial photographs that depict three different landslide sites on Lantau Island in Hong Kong, China. The bitemporal VHR images have a spatial resolution of 0.5 m/pixel. The pre- and postevent images for data sets A and B were, respectively, acquired on April 2007 and July 2014. Data set C was captured in 2006 and 2011. The size and ground reference of each data set are provided in Table I.

For data set D, a pair of images were collected by the Quick Bird satellite on April 2007 and February 2009. These image scenes covered an area for land use change in Jinan City, Shandong Province, China. The spatial resolution of the bitemporal VHR images was 0.6 m/pixel. The bitemporal VHR images depicted the land-use change event from farmland to building, as shown in Fig. 3.

Four evaluation indicators, including false alarm (FA), missed alarm (MA), total error (TE), and Kappa coefficient (Ka), which have been widely used to quantitatively evaluate the accuracy of BCDM [53], [54], were adopted for quantitative comparisons in the experiments. These indicators are computed as follows: 1) FA $=$ (FP/(TP + FN)); 2) MA $=$ (FN/(TP + FN)); and 3) TE $=$ ((FP + FN)/(TP + TN + FP + FN), where the true positive (TP) and true negative (TN) are the total number of correctly detected pixels for "changed" and "unchanged,"

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:21:44 UTC from IEEE Xplore. Restrictions apply.

TABLE II
PARAMETER SETTINGS OF DIFFERENT LCCD METHODS FOR EACH DATA SET

Method	Parameter Settings			
	Dataset-A	Dataset-B	Dataset-C	Dataset-D
CVA_FCM [29]	-	-	-	-
MLS [30]	L=4 μ=0.3	L=2 μ=0.3	L=4 μ=0.3	L=2 μ=0.1
LSELUC [31]	S=9	S=9	S=3	S=3
MAC_OOCD [4]	depth=5 area=3	depth=5 area=3	depth=5 area=3	depth=5 area=3
MLS_SVM [39]	scale=20 shape=0.8 compactness=0.9	scale=25 shape=0.8 compactness=0.9	scale=40 shape=0.8 compactness=0.9	scale=10 shape=0.8 compactness=0.9
Proposed approach	scale=20 shape=0.8 compactness=0.9	scale=25 shape=0.8 compactness=0.9	scale=40 shape=0.8 compactness=0.9	scale=10 shape=0.8 compactness=0.9

respectively; the false negative (FN) is the number of missed detected pixels by the method, and the false positive (FP) is the number of FA pixels. In addition, Ka is employed here to measure the interrater reliability for qualitative items of detection results, $Ka \in [0, 1]$; more details about Ka can be tracked in [55].

B. Experimental Setup

Five LCCD methods, including three widely used pixel-based methods and two relatively new object-oriented approaches, namely, CVA_FCM [29], MLS [30], LSELUC [31], MLS_SVM [39], and MAC_OOCD [4], were compared with the proposed KPVD-based change detection approach to investigate its feasibility and effect. In the following experiments, the parameters of each method were optimized by the trial-and-error method. The details of the optimization parameter setting for each data set are presented in Table II.

C. Experimental Results and Visual Identification

1) Data Sets A, B, and C: For data set A (Fig. 3), the spectral homogeneity of the bitemporal images is different. Detecting the landslide area with data set A is a challenging task because this area is almost covered by grass, and season considerably affects the spectral reflectance. The comparison of the results of different methods is shown in Table III. The object-oriented MAC_OOCD method obtained a better result compared with that of classical pixel-based methods in terms of FA and TE but had the worst accuracy in terms of MA. The proposed KPVD-based approach achieved the best accuracies for data set A in terms of FA, MA, TE, and Ka. The visual performance in Fig. 4 acquired by these different methods further verified the conclusion of quantitative comparisons. Based on visual comparison, the landslide inventory map acquired by the proposed KPVD-based approach had the least salt-and-pepper noise and presented the integral shape of the landslide. A conclusion similar to that of data set A can be obtained based on the results of data sets B and C in Tables IV and V, respectively. Furthermore, the visual performance for the data sets B and C, respectively, as shown in Figs. 5 and 6, further supported this quantitatively compared conclusion.

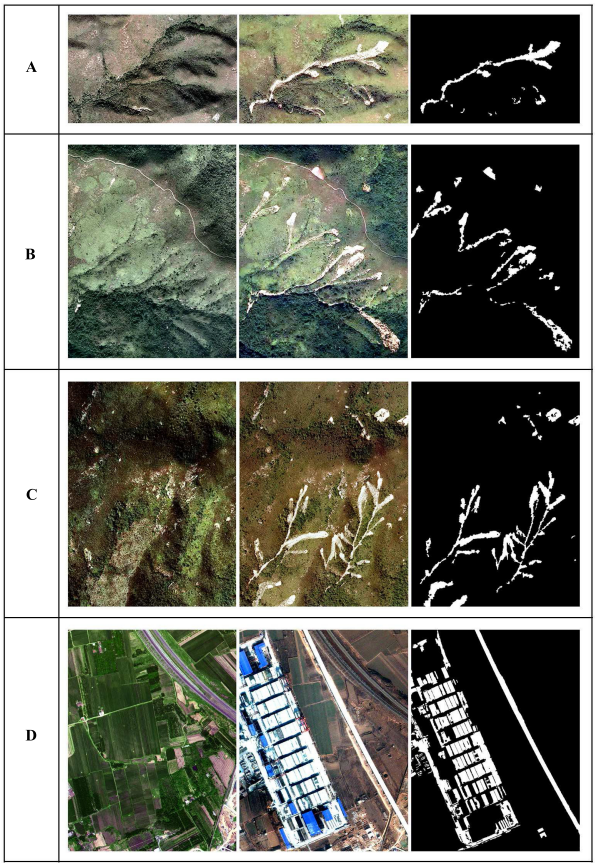


Fig. 3. Data set for each area and their reference maps. (From left to right) Data set label, pre-event images, postevent images, and ground reference maps.

2) Data Set D: The BCDM of data set D is obtained by the proposed KPVD approach and compared with that of the five methods, including CVA_FCM [29], MLS [30], LSELUC [31], MLS_SVM [39], and MAC_OOCD [4], to further investigate the feasibility and performance of the proposed KPVD-based change detection approach. The quantitative assessment of the results is presented in Table VI. The proposed approach achieved FA = 9.74%, MA = 3.70%, TE = 8.72%, and Ka = 0.911, which are the best accuracies among those of other methods. Furthermore, the proposed approach achieved

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:21:44 UTC from IEEE Xplore. Restrictions apply.

TABLE III

COMPARISON BETWEEN OTHER METHODS AND THE PROPOSED APPROACH FOR DATA SET A; THE ACCURACIES OF FA, MA, AND TE ARE PRESENTED IN PERCENTAGE (%), AND KA IS RANGED FROM 0 TO 1

Method	FA	MA	TE	Ka
CVA_FCM [29]	20.10	8.38	19.60	0.802
MLS [30]	9.40	6.85	9.30	0.906
LSELUC [31]	4.53	4.03	4.51	0.955
MAC_OOCD [4]	2.02	35.29	3.04	0.905
MLS_SVM [39]	9.79	0.57	9.34	0.906
Proposed approach	0.83	3.11	0.94	0.991

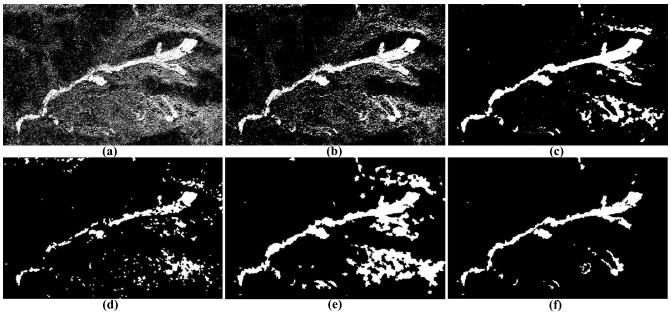


Fig. 4. BCDMs with different methods for data set A. (a) CVA_FCM [29]. (b) MLS [30]. (c) LSELUC [31]. (d) MAC_OOCD [4]. (e) MLS_SVM [39]. (f) Proposed KPVD approach.

TABLE IV

COMPARISON BETWEEN OTHER METHODS AND THE PROPOSED APPROACH FOR DATA SET B; THE ACCURACIES OF FA, MA, AND TE ARE PRESENTED IN PERCENTAGE (%), AND KA IS RANGED FROM 0 TO 1

Method	FA	MA	TE	Ka
CVA_FCM [29]	17.05	17.30	17.06	0.827
MLS [30]	6.53	16.40	7.02	0.929
LSELUC [31]	2.34	15.70	3.00	0.970
MAC_OOCD [4]	1.42	27.93	2.72	0.972
MLS_SVM [39]	5.19	0.42	4.95	0.950
Proposed approach	0.53	14.95	1.24	0.988

TABLE V

COMPARISON BETWEEN OTHER METHODS AND THE PROPOSED APPROACH FOR DATA SET C; THE ACCURACIES OF FA, MA, AND TE ARE PRESENTED IN PERCENTAGE (%), AND KA IS RANGED FROM 0 TO 1

Method	FA	MA	TE	Ka
CVA_FCM [29]	5.91	18.08	6.69	0.932
MLS [30]	2.34	14.27	3.11	0.969
LSELUC [31]	1.24	11.34	1.89	0.981
MAC_OOCD [4]	0.96	19.00	2.12	0.979
MLS_SVM [39]	1.14	3.62	1.30	0.987
Proposed approach	0.69	8.99	1.22	0.988

the best false detection while maintaining optimal missed detection and TE detection. The visual performance of these comparisons supports the quantitative conclusion for data set D, as shown in Fig. 7.

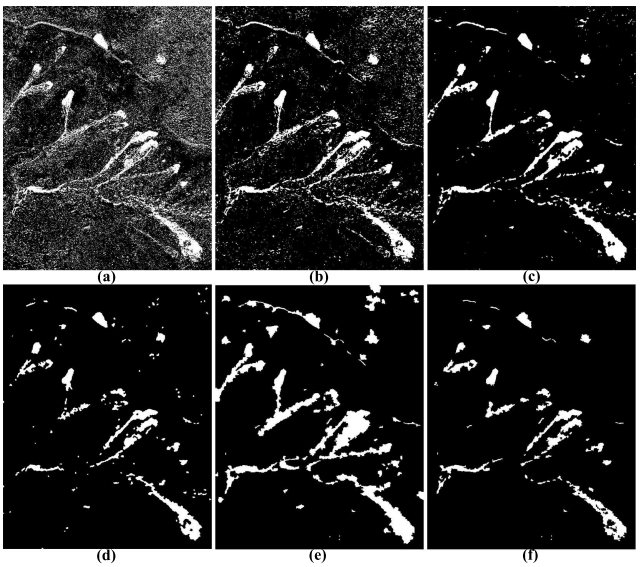


Fig. 5. BCDMs with different methods for data set B. (a) CVA_FCM [29]. (b) MLS [30]. (c) LSELUC [31]. (d) MAC_OOCD [4]. (e) MLS_SVM [39]. (f) Proposed KPVD approach.

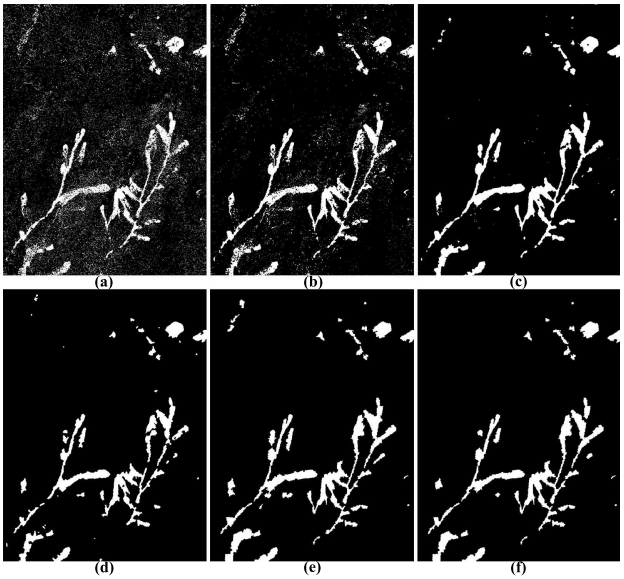


Fig. 6. BCDMs with different methods for data set C. (a) CVA_FCM [29]. (b) MLS [30]. (c) LSELUC [31]. (d) MAC_OOCD [4]. (e) MLS_SVM [39]. (f) Proposed KPVD approach.

Based on the experimental comparisons and analysis, as the bar comparisons in Figs. 8–11, it can be found that the proposed KPVD-based approach is effective and achieves the best accuracies compared with those of CVA_FCM [29], MLS [30], LSELUC [31], MLS_SVM [39], and MAC_OOCD [4], in terms of FA, MA, TE, and Ka.

D. Discussion

The aforementioned comparisons show that the proposed KPVD-based detection approach is better than the widely used CVA_FCM [29], MLS [30], LSELUC [31], MLS_SVM [39], and MAC_OOCD [4] in terms of quantitative accuracies and visual performance. Two of the four data sets are first selected in this section to demonstrate the relationship between

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:21:44 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

COMPARISON BETWEEN OTHER METHODS AND THE PROPOSED APPROACH FOR DATA SET D; THE ACCURACIES OF FA, MA, AND TE ARE PRESENTED IN PERCENTAGE (%), AND KA IS RANGED FROM 0 TO 1

Method	FA	MA	TE	Ka
CVA_FCM [29]	13.60	3.84	11.90	0.878
MLS [30]	9.75	5.10	8.97	0.908
LSELUC [31]	10.90	4.15	9.79	0.900
MAC_OOCD [4]	10.00	5.57	9.25	0.905
MLS_SVM [39]	11.70	3.66	10.34	0.894
Proposed approach	9.74	3.70	8.72	0.911

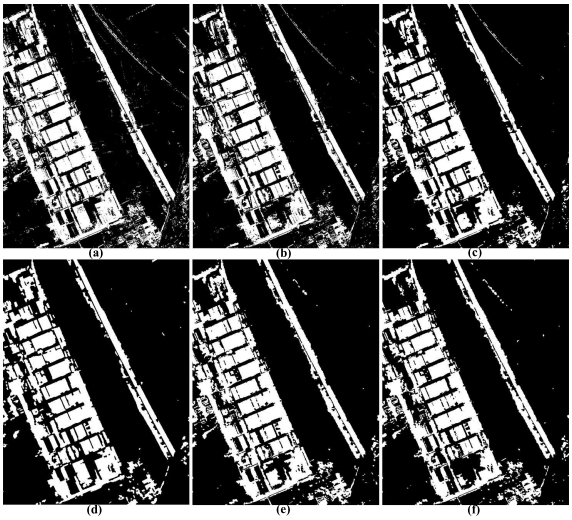


Fig. 7. BCDMs with different methods for data set D. (a) CVA_FCM [29]. (b) MLS [30]. (c) LSELUC [31]. (d) MAC_OOCD [4]. (e) MLS_SVM [39]. (f) Proposed KPVD approach.

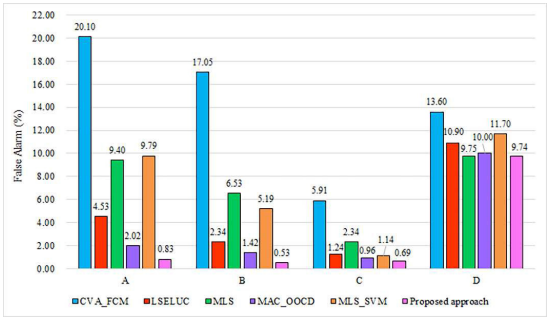


Fig. 8. FA accuracies with different LCCD methods based on each data set.

scale parameter and detection accuracy for the proposed KPVD-based detection approach to further promote the proposed KPVD approach in practical application. The intrinsic nature of the proposed KPVD is then discussed via comparison with the traditional pixel-to-pixel distance.

First, the sensitivity between the segmentation scale and the detection accuracies (FA, MA, and TE) is, respectively, presented in Figs. 12 and 13 for the proposed KPVD-based change detection approach. Fig. 12 for data set B shows that when the value of scale ranges from 10 to 25 with the fixed shape = 0.9 and compactness = 0.8, the accuracies reached the best performance of the proposed approach (FA = 0.53, MA = 1.23, and TE = 14.95). When scale is larger than 25,

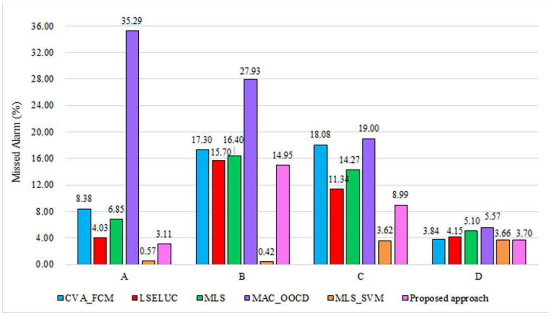


Fig. 9. MA accuracies with different LCCD methods based on each data set.

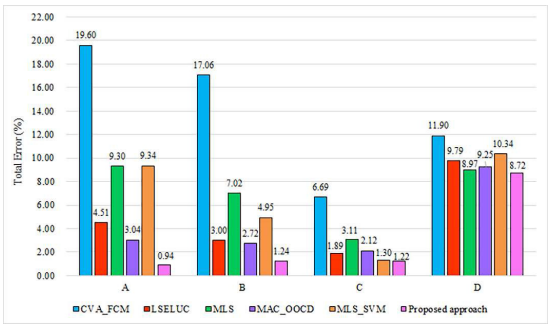


Fig. 10. TE accuracies with different LCCD methods based on each data set.

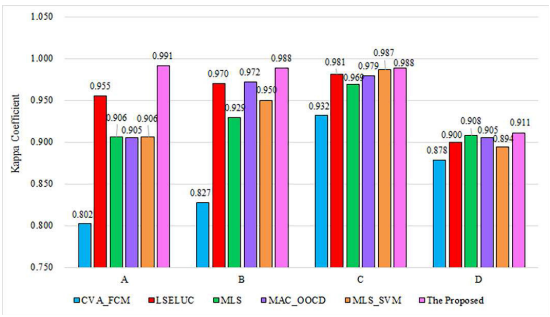


Fig. 11. Kappa coefficient with different LCCD methods based on each data set.

FA and TE are maintained with similar levels, but the value of MA is gradually increased. The reason for this finding is that a target may be under segmented when the value of scale is considerably large such that the shape of the change area cannot be objectively outlined in detail. The relationship of the parameter between the scale and detection accuracies for data set D shows that the detection accuracies are insensitive for the scale parameter variation, as shown in Fig. 13. Therefore, in the application of the proposed KPVD-based approach for detecting land cover change in real engineering, the following two aspects should be considered: 1) the segmental scale should be adjusted according to the data set, and the optimal scale for a specific image should be tested by trial-and-error experiments and 2) regardless of the missed detection, FA and TE are insensitive for the variation of the segmental scale parameter.

Second, a comparison between the KPVD and the traditional pixel-to-pixel or object-to-object distance is observed on the changed and unchanged areas (Fig. 14) to further demonstrate

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:21:44 UTC from IEEE Xplore. Restrictions apply.

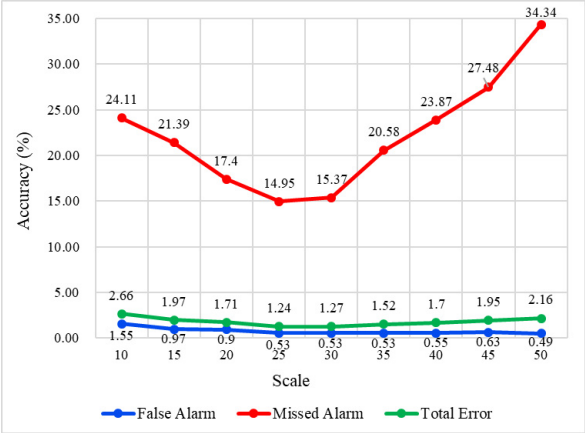


Fig. 12. Sensitivity analysis of the scale parameter for data set B.

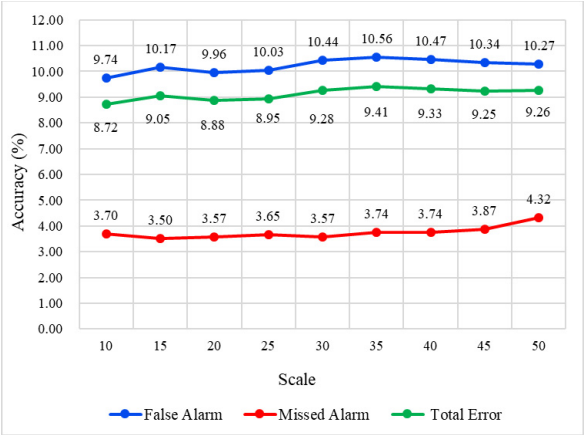


Fig. 13. Sensitivity analysis of the scale parameter for data set D.

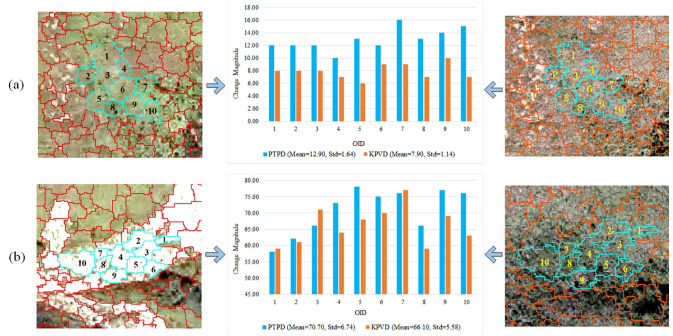


Fig. 14. Examples of the difference between the KPVD and the pixel-to-pixel distance in terms of standard deviation (Std.) and mean. (a) Unchanged and (b) changed area analyses.

the intrinsic nature of the proposed KPVD for measuring the change magnitude between an object area for the bitemporal images. Based on the comparison, the change magnitude based on the KPVD measurement posed a small standard deviation regardless of the changed or unchanged area. Thus, the proposed KPVD can improve homogeneity in terms of change magnitude. This improvement is helpful for determining a threshold to divide an area with high homogeneous magnitude into changed or unchanged area.

IV. Conclusion

In this article, a novel object-based distance, KPVD, is proposed for LCCD with VHR remote sensing images. First, a multiscale segmentation algorithm, FNEA, is employed to generate the multiscale objects. Second, key points are defined based on the box-whisper plot to reduce the negative affects on the LCCD from the outlier pixels, and key points of each object for the bitemporal images are extracted object by object. Finally, the change magnitude between the local areas of an object in the bitemporal images is calculated by the pairwise KPVs of the object in a band-to-band manner. CMI between the bitemporal images is generated when the bitemporal image is scanned and processed in this manner. Finally, the unsupervised Otsu binary threshold method is adopted to divide the CMI into a BCDM. The main contribution of the proposed approach lies in the novel distance that can measure the change magnitude between the areas of an object in the bitemporal for LCCD using VHR remote sensing images. Furthermore, the proposed KPVD is beneficial for inhibiting the noise of LCCD map.

The case studies based on three landslide sites and a land-use change event demonstrated that the proposed KPVD is effective and feasible for LCCD with VHR remote sensing images. Furthermore, KPVD can restrain the salt-and-pepper noise in the detection map and improve the detection accuracies compared with the widely used methods CVA_FCM [29], MLS [30], LSELUC [31], MLS_SVM [39], and MAC_OOCD [4]. The advantages and contributions of the proposed KPVD are summarized as follows.

1) Advanced change detection results are obtained with the proposed approach. For the two real land-cover change events with four pairs of VHR remote sensing images, the experimental results indicate that the proposed approach outperforms five widely used LCCD methods, namely, CVA_FCM [29], MLS [30], LSELUC [31], MLS_SVM [39], and MAC_OOCD [4], in terms of visual performance and three quantitative evaluation criteria.
2) Measuring the change magnitude between an object in the bitemporal images through the proposed KPVD is effective in inhibiting pseudo-change noise and improving detection accuracies. The spectral reflectance of the same ground target is usually different due to various acquired dates and imaging conditions of the bitemporal images, and this difference may bring pseudo changes in the detection map. However, according to the investigations and comparisons based on the four different change cases, the proposed KPVD is effective against the change of spectral reflectance varied by season or photo conditions during the KPVD application for change magnitude measurement. This finding is helpful for reducing the pseudo change of LCCD with VHR remote sensing images.
3) A simple no-parameter approach can be acceptable and easy to promote in the practical engineering application. In the proposed approach, apart from the parameters of the FNEA segmentation algorithm, the key points of an object are defined and extracted without any parameters. Moreover, the binary threshold-Otsu method is unsupervised yet effective for dividing the CMI into BCDM.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:21:44 UTC from IEEE Xplore. Restrictions apply.

Despite the advantages of the proposed KPVD-based approach, the practicality and widespread application of this approach should be further investigated. Moreover, two limitations of the proposed approach should be noted when applying it in practical applications: 1) multiscale segmentation should be conducted and this requires that the practitioner has a background in remote sensing and 2) the proposed approach only shows an improved performance for high-resolution remote sensing images because it is an object-based approach. In our future research, additional characteristics of the key points for an object will be extensively studied. Furthermore, additional applications based on these extended characteristics, such as land cover classification or target recognition, will be explored.

ACKNOWLEDGMENT

The authors are grateful to the Editor-in-Chief, Associate Editor, and reviewers for their insightful comments and suggestions.

REFERENCES

[1] A. Singh, "Review article digital change detection techniques using remotely-sensed data," *Int. J. Remote Sens.*, vol. 10, no. 6, pp. 989–1003, Jul. 1989.

[2] F. Luo, B. Du, L. Zhang, L. Zhang, and D. Tao, "Feature learning using spatial-spectral hypergraph discriminant analysis for hyperspectral image," *IEEE Trans. Cybern.*, vol. 49, no. 7, pp. 2406–2419, Jul. 2019.

[3] F. Luo, H. Huang, Y. Duan, J. Liu, and Y. Liao, "Local geometric structure feature for dimensionality reduction of hyperspectral imagery," *Remote Sens.*, vol. 9, no. 8, p. 790, Aug. 2017.

[4] L. Cai, W. Shi, H. Zhang, and M. Hao, "Object-oriented change detection method based on adaptive multi-method combination for remote-sensing images," *Int. J. Remote Sens.*, vol. 37, no. 22, pp. 5457–5471, Oct. 2016.

[5] P. Canuti, N. Casagli, L. Ermini, R. Fanti, and P. Farina, "Landslide activity as a geoindicator in italy: Significance and new perspectives from remote sensing," *Environ. Geol.*, vol. 45, no. 7, pp. 907–919, May 2004.

[6] B. Chen, Z. Chen, L. Deng, Y. Duan, and J. Zhou, "Building change detection with RGB-D map generated from UAV images," *Neurocomputing*, vol. 208, pp. 350–364, Oct. 2016.

[7] L. Dong and J. Shan, "A comprehensive review of earthquake-induced building damage detection with remote sensing techniques," *ISPRS J. Photogramm. Remote Sens.*, vol. 84, pp. 85–99, Oct. 2013.

[8] X.-L. Chen, H.-M. Zhao, P.-X. Li, and Z.-Y. Yin, "Remote sensing image-based analysis of the relationship between urban heat island and land use/cover changes," *Remote Sens. Environ.*, vol. 104, no. 2, pp. 133–146, Sep. 2006.

[9] M. K. Ridd and J. Liu, "A comparison of four algorithms for change detection in an urban environment," *Remote Sens. Environ.*, vol. 63, no. 2, pp. 95–100, Feb. 1998.

[10] T. Leichtle, C. Geiß, M. Wurm, T. Lakes, and H. Taubenböck, "Unsupervised change detection in VHR remote sensing imagery–an object-based clustering approach in a dynamic urban environment," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 54, pp. 15–27, 2017.

[11] M. A.-A. Hoque, S. Phinn, C. Roelfsema, and I. Childs, "Tropical cyclone disaster management using remote sensing and spatial analysis: A review," *Int. J. Disaster Risk Reduction*, vol. 22, pp. 345–354, Jun. 2017.

[12] R. Anniballe *et al.*, "Earthquake damage mapping: An overall assessment of ground surveys and VHR image change detection after L'Aquila 2009 earthquake," *Remote Sens. Environ.*, vol. 210, pp. 166–178, Jun. 2018.

[13] P. Lu, Y. Qin, Z. Li, A. C. Mondini, and N. Casagli, "Landslide mapping from multi-sensor data through improved change detection-based Markov random field," *Remote Sens. Environ.*, vol. 231, Sep. 2019, Art. no. 111235.

[14] K. S. Willis, "Remote sensing change detection for ecological monitoring in United States protected areas," *Biol. Conservation*, vol. 182, pp. 233–242, Feb. 2015.

[15] J. C. White, N. C. Coops, M. A. Wulder, M. Vastaranta, T. Hilker, and P. Tompalski, "Remote sensing technologies for enhancing forest inventories: A review," *Can. J. Remote Sens.*, vol. 42, no. 5, pp. 619–641, Jul. 2016.

[16] Z. Lv, T. Liu, P. Zhang, J. Atli Benediktsson, and Y. Chen, "Land cover change detection based on adaptive contextual information using bi-temporal remote sensing images," *Remote Sens.*, vol. 10, no. 6, p. 901, Jun. 2018.

[17] G. Jianya, S. Haigang, M. Guorui, and Z. Qiming, "A review of multi-temporal remote sensing data change detection algorithms," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 37, no. B7, pp. 757–762, 2008.

[18] A. P. Tewkesbury, A. J. Comber, N. J. Tate, A. Lamb, and P. F. Fisher, "A critical synthesis of remotely sensed optical image change detection techniques," *Remote Sens. Environ.*, vol. 160, pp. 1–14, Apr. 2015.

[19] X. Huang, Q. Lu, and L. Zhang, "A multi-index learning approach for classification of high-resolution remotely sensed images over urban areas," *ISPRS J. Photogramm. Remote Sens.*, vol. 90, pp. 36–48, Apr. 2014.

[20] M. Li, S. Zang, B. Zhang, S. Li, and C. Wu, "A review of remote sensing image classification techniques: The role of spatio-contextual information," *Eur. J. Remote Sens.*, vol. 47, no. 1, pp. 389–411, Feb. 2017.

[21] C. Huo, K. Chen, K. Ding, Z. Zhou, and C. Pan, "Learning relationship for very high resolution image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 8, pp. 3384–3394, Aug. 2016.

[22] M. Hussain, D. Chen, A. Cheng, H. Wei, and D. Stanley, "Change detection from remotely sensed images: From pixel-based to object-based approaches," *ISPRS J. Photogramm. Remote Sens.*, vol. 80, pp. 91–106, Jun. 2013.

[23] T. Celik, "Unsupervised change detection in satellite images using principal component analysis and k-means clustering," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 772–776, Aug. 2009.

[24] B. P. Salmon, J. C. Olivier, K. J. Wessels, W. Kleynhans, F. van den Bergh, and K. C. Steenkamp, "Unsupervised land cover change detection: Meaningful sequential time series analysis," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 4, no. 2, pp. 327–335, Jun. 2010.

[25] P. Zhang, Z. Lv, and W. Shi, "Local spectrum-trend similarity approach for detecting land-cover change by using SPOT-5 satellite images," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 4, pp. 738–742, Apr. 2013.

[26] M. Volpi, D. Tuia, F. Bovolo, M. Kanevski, and L. Bruzzone, "Supervised change detection in VHR images using contextual information and support vector machines," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 20, pp. 77–85, Feb. 2013.

[27] L. ZhiYong, W. Shi, J. A. Benediktsson, and L. Gao, "A modified mean filter for improving the classification performance of very high-resolution remote-sensing imagery," *Int. J. Remote Sens.*, vol. 39, no. 3, pp. 770–785, Oct. 2018.

[28] Z. Lv, T. Liu, C. Shi, J. A. Benediktsson, and H. Du, "Novel land cover change detection method based on k-means clustering and adaptive majority voting using bitemporal remote sensing images," *IEEE Access*, vol. 7, pp. 34425–34437, 2019.

[29] S. Liu, L. Bruzzone, F. Bovolo, M. Zanetti, and P. Du, "Sequential spectral change vector analysis for iteratively discovering and detecting multiple changes in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 8, pp. 4363–4378, Aug. 2015.

[30] Y. Bazi, F. Melgani, and H. D. Al-Sharari, "Unsupervised change detection in multispectral remotely sensed imagery with level set methods," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 8, pp. 3178–3187, Aug. 2010.

[31] X. Zhang, W. Shi, P. Liang, and M. Hao, "Level set evolution with local uncertainty constraints for unsupervised change detection," *Remote Sens. Lett.*, vol. 8, no. 8, pp. 811–820, May 2017.

[32] T. Blaschke, "Object based image analysis for remote sensing," *ISPRS J. Photogramm. Remote Sens.*, vol. 65, no. 1, pp. 2–16, Jan. 2010.

[33] T. Blaschke *et al.*, "Geographic object-based image analysis–towards a new paradigm," *ISPRS J. Photogramm. Remote Sens.*, vol. 87, pp. 180–191, 2014.

[34] V. Walter, "Object-based classification of remote sensing data for change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 58, nos. 3–4, pp. 225–238, Jan. 2004.

[35] B. Desclée, P. Bogaert, and P. Defourny, "Forest change detection by statistical object-based method," *Remote Sens. Environ.*, vol. 102, nos. 1–2, pp. 1–11, May 2006.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:21:44 UTC from IEEE Xplore. Restrictions apply.

[36] G. Chen, G. J. Hay, L. M. Carvalho, and M. A. Wulder, "Object-based change detection," *Int. J. Remote Sens.*, vol. 33, no. 14, pp. 4434–4457, 2012.
[37] S. Bontemps, P. Bogaert, N. Titeux, and P. Defourny, "An object-based change detection method accounting for temporal dependences in time series with medium to coarse spatial resolution," *Remote Sens. Environ.*, vol. 112, no. 6, pp. 3181–3191, Jun. 2008.
[38] C. Zhang, G. Li, and W. Cui, "High-resolution remote sensing image change detection by statistical-object-based method," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 7, pp. 2440–2447, Jul. 2018.
[39] G. Cao, Y. Li, Y. Liu, and Y. Shang, "Automatic change detection in high-resolution remote-sensing images by means of level set evolution and support vector machine classification," *Int. J. Remote Sens.*, vol. 35, no. 16, pp. 6255–6270, Aug. 2014.
[40] F. Deng, C. Yang, C. Cao, and X. Fan, "An improved method of FNEA for high resolution remote sensing image segmentation," *J. Geo-Inf. Sci*, vol. 16, pp. 95–101, 2014.
[41] Y. Yang, H. Li, Y. Han, and H. Gu, "High resolution remote sensing image segmentation based on graph theory and fractal net evolution approach," *Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci*, vol. 40, pp. 197–201, 2014.
[42] L. Dingle Robertson and D. J. King, "Comparison of pixel-and object-based classification in land cover change mapping," *Int. J. Remote Sens.*, vol. 32, no. 6, pp. 1505–1529, 2011.
[43] X. Zhang, P. Xiao, X. Song, and J. She, "Boundary-constrained multi-scale segmentation method for remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 78, pp. 15–25, Apr. 2013.
[44] U. C. Benz, P. Hofmann, G. Willhauck, I. Lingenfelder, and M. Heynen, "Multi-resolution, object-oriented fuzzy analysis of remote sensing data for GIS-ready information," *ISPRS J. Photogramm. Remote Sens.*, vol. 58, nos. 3–4, pp. 239–258, Jan. 2004.
[45] M. Baatz and A. Schape, "Multiresolution segmentation: An optimization approach for high quality multi-scale image segmentation," in *Proc. Angew. Geographische Informations-Verarbeitung XII*, 2000, pp. 12–13.
[46] H. T. Li, H. Y. Gu, Y. S. Han, and J. H. Yang, "An efficient multi-scale segmentation for high-resolution remote sensing imagery based on statistical region merging and minimum heterogeneity rule," in *Proc. Int. Workshop Earth Observ. Remote Sens. Appl.*, Jun. 2008, pp. 1–6.
[47] M.-T. Pham, G. Mercier, and J. Michel, "Pointwise graph-based local texture characterization for very high resolution multispectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 5, pp. 1962–1973, May 2015.
[48] Z. Lv, T. Liu, J. Atli Benediktsson, T. Lei, and Y. Wan, "Multi-scale object histogram distance for LCCD using bi-temporal Very-High-Resolution remote sensing images," *Remote Sens.*, vol. 10, no. 11, p. 1809, Nov. 2018.
[49] H. Zhuang, Z. Tan, K. Deng, and G. Yao, "Change detection in multispectral images based on multiband structural information," *Remote Sens. Lett.*, vol. 9, no. 12, pp. 1167–1176, Oct. 2018.
[50] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Trans. Syst., Man, Cybern.*, vol. SMC-9, no. 1, pp. 62–66, Jan. 1979.
[51] M. V. Herbei, V. Ciolac, A. Şmuleac, E. Nistor, and L. Ciolac, "Geo-referencing of topographical maps using the software ARCGIS," *Res. J. Agricult. Sci.*, vol. 42, no. 3, pp. 595–606, 2010.
[52] T. Nelson, H. G. Wilson, B. Boots, and M. A. Wulder, "Use of ordinal conversion for radiometric normalization and change detection," *Int. J. Remote Sens.*, vol. 26, no. 3, pp. 535–541, Feb. 2005.
[53] Z. Lv, T. Liu, Y. Wan, J. A. Benediktsson, and X. Zhang, "Post-processing approach for refining raw land cover change detection of very high-resolution remote sensing images," *Remote Sens.*, vol. 10, no. 3, p. 472, Mar. 2018.
[54] L. Bruzzone and D. F. Prieto, "Automatic analysis of the difference image for unsupervised change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 3, pp. 1171–1182, May 2000.
[55] M. L. McHugh, "Interrater reliability: The kappa statistic," *Biochem. Medica*, vol. 22, no. 3, pp. 276–282, 2012.



Zhiyong Lv received the M.S. and Ph.D. degrees from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in 2008 and 2014, respectively.

He was an Engineer of surveying and worked in the First Institute of Photogrammetry and Remote Sensing from 2008 to 2011. He is currently with the School of Computer Science and Engineering, Xi'an University of Technology, Xi'an, China. His research interests include multihyperspectral and high-resolution remotely sensed image processing, spatial feature extraction, neural networks, pattern recognition, deep learning, and remote sensing applications.



Tongfei Liu is currently pursuing the master's degree in computer science with the Xi'an University of Technology, Xi'an, China.

He is interested in spatial–spectral feature extraction, pattern recognition, ground target detection, and land cover/land use change detection, through remote sensing image with high or very high spatial resolution (including satellite imagery and aerial images).



Jón Atli Benediktsson (Fellow, IEEE) received the Cand.Sci. degree in electrical engineering from the University of Iceland, Reykjavik, Iceland, in 1984, and the M.S.E.E. and Ph.D. degrees from Purdue University, West Lafayette, IN, USA, in 1987 and 1990, respectively.

He is currently a Pro-Rector for Academic Affairs and a Professor of electrical and computer engineering with the University of Iceland. His research interests are in remote sensing, biomedical analysis of signals, pattern recognition, image processing, and signal processing, and he has published extensively in these fields.

Dr. Benediktsson is a fellow of SPIE. He is a member of the Association of Chartered Engineers in Iceland (VFI), the Societas Scinetiarum Islandica, and Tau Beta Pi. He was the 2011-2012 President of the IEEE Geoscience and Remote Sensing Society (GRSS) and has been on the GRSS AdCom since 2000. He received the Stevan J. Kristof Award from Purdue University in 1991 as an Outstanding Graduate Student in remote sensing, the Icelandic Research Council's Outstanding Young Researcher Award in 1997, the IEEE Third Millennium Medal in 2000, the Yearly Research Award from the Engineering Research Institute, University of Iceland, in 2006, the Outstanding Service Award from the IEEE Geoscience and Remote Sensing Society in 2007, and the IEEE/VFI Electrical Engineer of the Year Award in 2013. He was co-recipient of the University of Iceland's Technology Innovation Award in 2004, the 2012 IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING Paper Award, and the IEEE GRSS Highest Impact Paper Award in 2013. He was an Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS) from 2003 to 2008 and has been serving as an Associate Editor for TGRS since 1999, the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS since 2003, and the IEEE ACCESS since 2013. He is on the International Editorial Board of the *International Journal of Image and Data Fusion* and was the Chairman of the Steering Committee of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (J-STARS) from 2007 to 2010. He is a Co-Founder of the biomedical startup company Oxymap (www.oxymap.com).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:21:44 UTC from IEEE Xplore. Restrictions apply.